Occultation geometry

Where does the measurement happen, and over what volume?

`Notebook companion: Unit-05-geometry.ipynb`

1 The question

Unit 4 established which wavelengths separate the species. This unit is about where along the line of sight the measurement is actually made, and how large a piece of atmosphere it averages over.

The question is not academic. The benchmark asks for a horizontal resolution of 5 degrees, and for any solar occultation that number is set by geometry rather than by design. The same geometry determines how much gas the ray passes through, and that in turn determines which altitudes give an optical depth large enough to measure. It also determines how often a profile is available, and how long the instrument has to collect each altitude bin. Both of those last two are properties of the host's orbit rather than of the instrument.

2 Physical picture

The instrument observes the Sun as it sets behind the Earth's limb. Early in an orbit the Sun is well above the host's local horizontal, and the line of sight to it passes well above the atmosphere. Later in the orbit the Sun drops towards the limb, so the line of sight begins to pass through the atmosphere. It continues to descend until the Earth blocks the Sun entirely.

Along any one line of sight there is a point of closest approach to the centre of the Earth. That point is the *tangent point*, and its height above the surface is the *tangent altitude* z_t. It is the natural label for a measurement, because most of the ray's path lies near that height.

One geometric fact constrains everything that follows. The tangent point is the closest approach of the line of sight to the centre of the Earth, and the host sits on that same line, so the tangent radius can never be larger than the host's orbital radius. **The highest tangent altitude the instrument can sample is the altitude of its own host.** A payload at 500 km can never observe a tangent point at 600 km.

Three distinct paths appear in this unit and they are easy to confuse.

The vertical column above an altitude is what a density profile is usually reported against. Nothing measures it directly. It appears here only as the denominator of a ratio.

The occulting ray enters the atmosphere, passes its tangent point, and leaves. This is what the instrument measures, and `occ.slant` computes it.

The reference ray is taken before the occultation begins, while the Sun is still above the host's local horizontal. That path never descends below the host. It is what supplies the unattenuated signal I_0, and section 4d shows that it is not quite unattenuated.

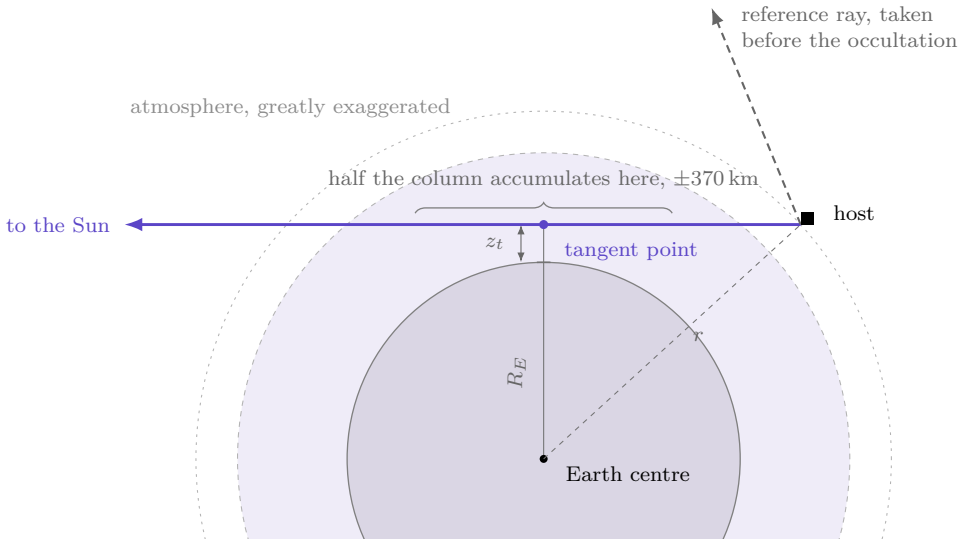


Figure 1: The occulting geometry, and the two rays that matter. Nothing here is to scale: the Earth's radius is 6371 km and a typical host orbits at 6871 km from the centre, a difference of eight per cent, and the atmosphere that does the absorbing is a few hundred kilometres thick. Drawn to scale, the ray, the tangent point and the host would all lie within the thickness of the line marking the surface. The brace marks the span computed in section 4b, not a fixed property of the geometry.

3 Derivation

Take the tangent point as the origin of path distance x along the ray. The ray is straight, so the altitude at path distance x follows from Pythagoras,

$$z(x) \;=\; \sqrt{(R_E + z_t)^2 + x^2} - R_E \;\simeq\; z_t + \frac{x^2}{2(R_E + z_t)},$$

where the expansion holds while $x \ll R_E$. The altitude therefore rises *quadratically* with distance from the tangent point, which means it rises slowly at first. That is why the limb geometry is useful. The ray stays close to its lowest altitude over a long stretch of its path, and that is the altitude at which the gas along the path is densest.

Now put that altitude into an atmosphere whose density falls exponentially with scale height H. The density along the ray becomes a Gaussian in path distance,

$$n(x) \;=\; n(z_t)\,\exp\!\left(-\frac{x^2}{2H(R_E + z_t)}\right),$$

and integrating it over the whole ray gives a slant column larger than the vertical column above the same altitude by

$$\frac{N_{\text{slant}}}{N_{\text{vertical}}} \;\simeq\; \sqrt{\frac{2\pi(R_E + z_t)}{H}}.$$

Two consequences follow, and they are two aspects of one fact. The enhancement is large, because R_E is much larger than H. It is larger still for a heavier species, because a heavier species has a smaller scale height, and a smaller scale height concentrates that species' density more tightly

around the tangent point. The same concentration that raises the enhancement also narrows the piece of atmosphere that the measurement averages over.

`occ.slant` does not use this expansion. It integrates the interpolated profile along the true path, so the analytic result above is a check on the numerical one rather than a substitute for it.

4 Worked numbers

4a. How much more gas a limb path traverses

z_t	$N_{\mathrm{slant}}(\mathrm{O})$	$N_{\mathrm{vert}}(\mathrm{O})$	O	N_2	O_2	H_{O}	analytic
100	3.91×10^{19}	7.20×10^{17}	54.4	83.2	89.6	12.7	57.1
150	2.59×10^{18}	8.06×10^{16}	32.1	43.0	48.4	29.8	37.1
200	7.49×10^{17}	2.52×10^{16}	29.8	39.6	42.6	43.1	31.0
250	2.57×10^{17}	8.78×10^{15}	29.3	38.9	41.6	47.2	29.7
300	9.22×10^{16}	3.17×10^{15}	29.1	38.6	41.3	48.7	29.3
400	1.25×10^{16}	4.34×10^{14}	28.9	38.3	41.0	50.4	28.9

Columns in cm^{-2}, altitudes and scale heights in km. The three middle columns are the ratio of slant to vertical column for each species. The analytic column evaluates $\sqrt{2\pi(R_E + z_t)/H_{\mathrm{O}}}$ and applies to atomic oxygen.

A ray at 250 km passes through about thirty times as much atomic oxygen as a vertical path from the same altitude, and about forty times as much molecular oxygen. This lifts the optical depth above the detector's stability floor, derived in Unit 6, at altitudes where a vertical path would give a value too small to measure.

The analytic formula agrees with the numerical integration to within two per cent above 250 km. It is worse lower down, by twenty per cent at 150 km and five per cent at 100 km, because the derivation assumed a single scale height and the real scale height changes rapidly across the lower thermosphere. Use the formula to reason and the code to compute.

4b. How wide a piece of atmosphere one ray averages

The slant column is an integral along a path more than a thousand kilometres long, so calling it a measurement *at* the tangent point is an approximation. How good that approximation is can be stated by asking what angular span, centred on the tangent point, contains a stated fraction of the column.

The fraction chosen has to be quoted alongside the answer. The distribution of the column along the ray has long tails, so the span changes a great deal with the fraction enclosed.

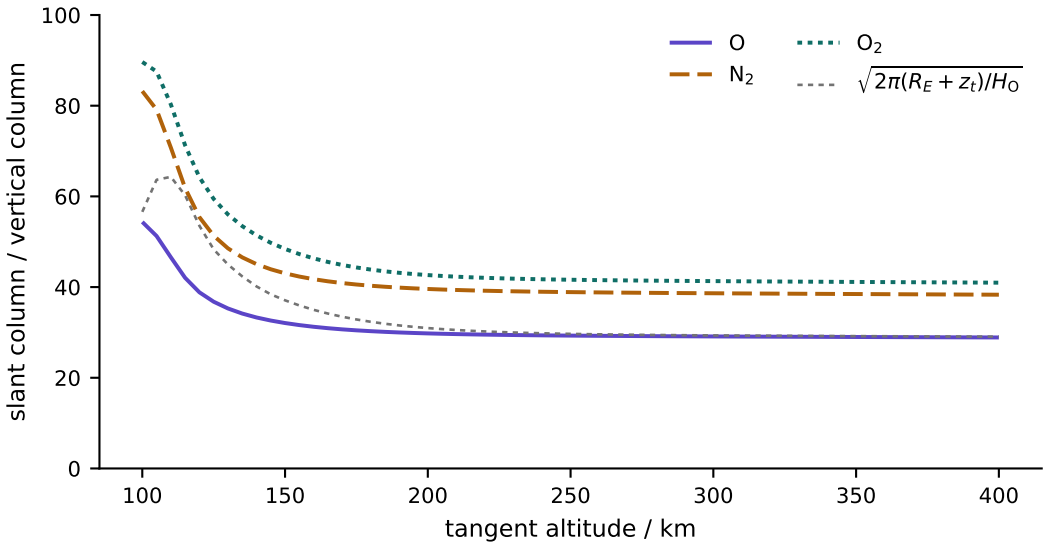


Figure 2: The limb enhancement against tangent altitude, for the three major species, with the analytic estimate for atomic oxygen shown for comparison. The enhancement rises towards low altitudes because the scale height falls there, and it is larger for heavier species for the same reason. Generated by section B of the notebook.

Atomic oxygen	25 %	50 %	75 %	90 %
200 km, quiet	3.05	6.48	11.11	15.97
250 km, quiet	3.12	6.62	11.30	16.18
300 km, quiet	3.15	6.67	11.37	16.27
250 km, ap = 400	3.60	7.65	13.12	18.89
300 km, ap = 400	3.68	7.81	13.35	19.14

Full angular width in degrees, centred on the tangent point, containing the stated fraction of the slant column.

Taking the central half of the column as the working definition, one ray through atomic oxygen above 250 km averages over 6.6 to 6.7 degrees in a quiet atmosphere and 7.7 to 7.9 degrees in a severe storm. Nitrogen is narrower, at 4.9 degrees at 200 km and 5.0 at 300 km, and molecular oxygen narrower still at 4.5 to 4.7 degrees, for the same reason that their limb enhancements are larger.

Against a benchmark of 5 degrees, the oxygen span is about a third wider than the benchmark allows and the nitrogen span very nearly meets it. This is intrinsic to the measurement. It follows from the curvature of the Earth and the scale height of the atmosphere, and no instrument measuring along a limb path can avoid it.

The project's documents have quoted a figure of 6.8 to 7.1 degrees above 250 km without recording either the enclosed fraction or the state of the atmosphere it assumed. That range falls between the quiet and storm values computed here, so it is not wrong, but it cannot be defended as written. Section 6 replaces it.

4c. The occultation is a timed event

The tangent altitude is not a setting. It descends on its own as the host moves, and two orbital quantities govern what the instrument can do about it.

The first is the *beta angle*, the angle between the direction to the Sun and the plane of the orbit. The lowest tangent altitude reached anywhere in an orbit is $r \sin \beta - R_E$, so beta determines how deep the profile reaches. Above $\beta = \arcsin(R_E/r)$ the host never enters the Earth's shadow at all, and there is no occultation to observe. That limit is 68.9 degrees for a host at 456 km, 68.0 degrees at 500 km and 66.1 degrees at 600 km.

The second is the descent rate, which sets how long the tangent point takes to cross each altitude bin. For a host at 500 km, the tangent point falls through 200 km at 2.23 km per second when the Sun lies in the orbital plane, and at 0.71 km per second at a beta angle of 65 degrees. A 10 km bin is therefore crossed in between 4.5 and 14 seconds. The descent from 300 to 100 km takes 91 seconds with the Sun in the plane, and longer as beta rises.

Those two effects work in opposite directions. In a grazing occultation the tangent point takes about three times as long to cross each bin, so about three times as many photons are collected, but the profile does not reach as deep. A profile is available at each sunrise and each sunset, so roughly two per orbit and about thirty per day.

Beta is a property of the host's orbit and of the season, not something the payload controls, and it turns into a host-selection criterion. A dawn–dusk sun-synchronous orbit keeps the Sun close to the orbit normal, which places beta near 90 degrees and produces no occultations at all for much of the year. The team brief describes a sun-synchronous host as a bonus. That is true for the cross-track pointing argument, and false for the beta-angle argument made here. Section 6 states the constraint.

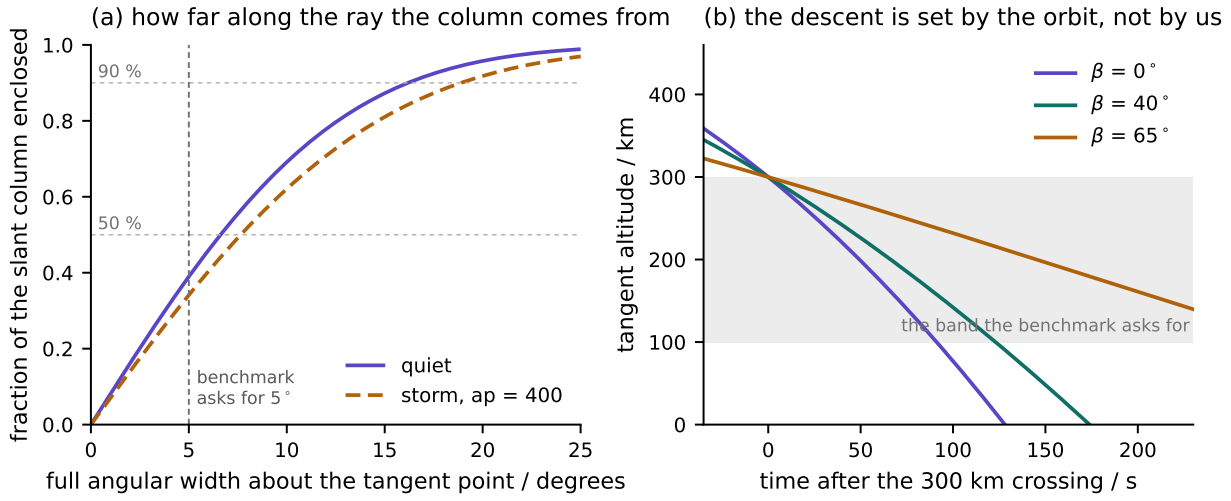


Figure 3: (a) The fraction of the slant column of atomic oxygen enclosed within a given angular span about the tangent point, at a tangent altitude of 250 km, for a quiet and a severely disturbed atmosphere. The dashed vertical line marks the 5 degree benchmark. (b) Tangent altitude against time for a host at 500 km at three beta angles, with the altitude band the benchmark asks for shaded. Both panels are generated by sections C and D of the notebook.

4d. The Sun is not a point, and I_0 is not exactly one

The solar disc subtends 0.53 degrees, or 9.3 mrad. Seen from a host at 500 km, the tangent point at 200 km lies 2008 km away, so that angular spread covers 18.7 km of tangent altitude at any instant. One binned detector row subtends 4.7 mrad, which is 10 km at the tangent point, so the disc covers about two rows at once. The instrument receives part of a profile immediately and the descent supplies the rest. This is the geometric reason the 3 September decision went to an imaging sensor. A detector that resolves the disc measures several tangent altitudes at the same instant; a detector that does not resolve it measures only their sum.

The reference frames are a different matter. When the Sun sits on the host's local horizontal, the line of sight to it still runs a long way through the atmosphere above the host, and at the extreme-ultraviolet wavelengths that atmosphere absorbs a small part of the light. The residual optical depth in the frames that are supposed to define I_0 is:

Solar elevation	456 km host	500 km	600 km
0° (on the horizontal)	0.0188	0.0080	0.0012
5°	0.0098	0.0042	0.0006
10°	0.0063	0.0027	0.0004
20°	0.0036	0.0015	0.0002

Residual optical depth of the aluminium channel in the reference geometry, quiet atmosphere. The zirconium and indium channels are smaller by a factor of two to four, and the far-ultraviolet channels are below 10^{-4} everywhere in this table.

An optical depth τ_0 left unmodelled in the reference frames subtracts directly from every measured optical depth, and so biases every slant column low by roughly τ_0/τ. For the lowest permitted host and a reference taken on the horizontal, that is nearly two per cent at $\tau = 1$, which is not negligible against an accuracy target of ten per cent. Raising the reference to 10 degrees of solar elevation cuts it to 0.6 per cent, and a host 150 km higher cuts it by a further factor of five.

5 What the computation shows

The limb geometry is what makes the measurement possible. A ray at 250 km passes through thirty times as much oxygen as a vertical path from the same altitude, so its optical depth is thirty times larger. The detector's stability floor, $\sigma_\tau = 0.014$, does not shrink when more light is collected, so a vertical path with an optical depth near 0.01 could not be measured with any aperture. Photon noise alone would need between about 130 and 740 times the collecting area, depending on the optical depth, to match the fractional error of the limb path.

Horizontal resolution is a definition before it is a number. The span containing half the column and the span containing ninety per cent of it differ by a factor of two and a half, and both are correct answers to loosely worded versions of the same question. Any figure we publish has to carry its enclosed fraction, its species and the state of the atmosphere it assumed.

The host's orbit sets both the cadence and the integration time. Beta determines how deep each profile reaches, whether an occultation happens at all, and how many seconds the instrument

has per altitude bin. None of those is a payload parameter, so all of them belong in the host interface requirements rather than in the instrument design.

6 What we decided

We replace the horizontal-resolution figure in the project's documents. The statement becomes: one ray averages over the central 50 per cent of its slant column, which spans 6.6 to 6.7 degrees for atomic oxygen above 250 km in a quiet atmosphere, 7.7 to 7.9 degrees in a severe storm, and about 5.0 degrees for molecular nitrogen. The benchmark asks for 5 degrees. We state the definition alongside the number every time it appears.

The host requirement gains a beta-angle constraint. A candidate orbit must spend a useful fraction of the year below about 66 to 69 degrees of beta, depending on its altitude, or it will produce no occultations. This excludes dawn–dusk sun-synchronous orbits and admits mid-morning ones.

Reference frames are taken with the Sun at least 10 degrees above the host's local horizontal, and the residual optical depth in them is modelled rather than assumed to be zero. The correction needs the host's position and the pointing direction, which is one of the reasons the host must supply GNSS position and time.

Refraction is not modelled anywhere in the forward model. Section 7 gives the number that justifies leaving it out.

7 What we do not claim

Not that the atmosphere is spherically symmetric, only that we have assumed it is. `occ.slant` integrates a density that depends on altitude alone. Section 4b shows the ray averaging over six or seven degrees of arc, and every occultation we observe happens at the day–night terminator, which is where the horizontal gradients in the thermosphere are steepest. The assumption is therefore worst exactly where we always measure. This is the fourth of the project's standing caveats, it is not repaired anywhere in this series, and Unit 8 includes its consequences in the error budget.

Not that I_0 is unattenuated. Section 4d gives the residual, which reaches 1.9 per cent in the aluminium channel for the lowest host altitude the project permits. We correct it; we do not ignore it.

Not that refraction is zero. Two things bend the ray. The neutral gas has a refractivity that scales with number density, and at the total density of $8.2 \times 10^{9}\,\mathrm{cm^{-3}}$ found at 200 km it is 9.0×10^{-14}, obtained by scaling the standard visible refractivity of air [1]. The free electrons of the ionosphere contribute a refractivity of the opposite sign that grows with wavelength; at an electron density of $10^{6}\,\mathrm{cm^{-3}}$ it is -4.0×10^{-13} at 30 nm and -1.1×10^{-11} at 155 nm. Taking the largest of these, the accumulated bending along a limb path is 3.4×10^{-10} radians. One 10 km tangent-altitude bin subtends 4.7×10^{-3} radians. The bending is smaller than the instrument's sampling by a factor of ten million. For comparison, radio occultation at GPS frequencies works in a regime where the bending is of order 10^{-2} radians and is the measurement [2].

Not that the model atmosphere extends far enough. Every path integral here is truncated at 800 km, and helium, which dominates the neutral density above roughly 400 km, is omitted from

the model entirely. Both matter more for the reference geometry of section 4d than for the occulting rays, because the reference path runs outward through exactly the region the model handles worst.

Not that the tangent point holds still. It does when the Sun lies in the orbital plane, and section 4c assumes it throughout. At other beta angles the tangent point drifts across the Earth during the descent, so a profile is assembled from slightly different locations at different altitudes. Exercise E5.2 estimates the size of that drift; we have not folded it into the error budget.

8 Exercises

E5.1 Recompute the limb enhancement for a host over the winter pole during a severe storm, using `occ.msis(lat=-70., ap=400.)`. The scale height rises. Predict from the analytic formula whether the enhancement rises or falls, and by roughly how much, before you run the calculation. Then explain the sign of the effect in terms of where along the ray the gas sits.

E5.2 Section 4c treats the tangent point as stationary. Show that this is exact when the Sun lies in the orbital plane, by writing the tangent point as the component of the host position perpendicular to the Sun direction and showing that its direction does not change. Then estimate how far the tangent point moves during a descent from 300 to 100 km at a beta angle of 40 degrees, and say whether that distance is large compared with the six-degree span of section 4b.

E5.3 Using the table in section 4d, choose a host altitude and a minimum solar elevation for the reference frames such that the resulting bias in the aluminium channel stays below 0.5 per cent of a slant column measured at an optical depth of 1. State what your answer assumes about how well the pointing direction is known, and what would happen to the bias if the pointing knowledge were ten times worse.

References

[1] P. E. Ciddor, "Refractive index of air: new equations for the visible and near infrared," *Appl. Opt.* **35**(9), 1566–1573 (1996), doi:10.1364/AO.35.001566.

[2] E. R. Kursinski, G. A. Hajj, J. T. Schofield, R. P. Linfield and K. R. Hardy, "Observing Earth's atmosphere with radio occultation measurements using the Global Positioning System," *J. Geophys. Res.* **102**(D19), 23429–23465 (1997), doi:10.1029/97JD01569.

[3] R. K. Sewell et al., "Thermospheric composition and temperature from GOES-R/SUVI solar occultations," *Space Weather* **23**, e2024SW004234 (2025); arXiv:2508.10242.

[4] G. R. Smith and D. M. Hunten, "Study of planetary atmospheres by absorptive occultations," *Rev. Geophys.* **28**(2), 117–143 (1990), doi:10.1029/RG028i002p00117.

[5] E. M. B. Thiemann et al., "Neutral density profiles from PROBA2/LYRA solar occultations," *Space Weather* **15**, 1649 (2017); arXiv:1708.08833.

Every quantitative result in this unit is provisional in the four ways that apply to the whole project: filter bands are idealized top-hats rather than measured transmission curves; the truth atmosphere is NRLMSIS 2.1; helium is omitted above 400 km; and no horizontal structure along the ray is modelled. Analysis code is in `channel-plan/`.